What does it mean? 数据说明了什么?

Pooled class results are stronger evidence than one group; the right graph shows the pattern, and CER plus an honest evaluation say how sure we are.

- I can choose the right graph and write a conclusion: claim, number evidence, science reasoning.
- I can explain one problem with our test, how it affected the results, and how sure we are.

pool (data) 汇总（数据） **class mean 全班平均值** **bar chart 条形图** **line graph 折线图** **claim 主张** **evidence 证据** **reasoning 推理** **evaluate 评价** **improvement 改进** **confident 有把握的**

THE LESSON

Do Now	4	Folders back; finish Week 8 group means (no anomalies), check against class table; Today slide
Hook	3	'Snow Owls 292 ml, Ice Foxes 300 ml: who is right?' Pair talk: trust one group or the class?
Learn	7	Pooling (class mean = mean of group means); bar or line vote; teacher models CER; Core A and B
Do	18	Panels by role: Core C, Stretch, Challenge, panel card, glue on poster; first glued by min 12
Think and Reflect	8	Self-assessment slip (4 min); split the Week 10 talk with talk frame card; exit ticket; collect

Hands-on: No practical: groups pool class data, choose a graph, and write, cut out and glue panels onto an A3 poster.

Collect or assess: Poster: evidence for strands 1 to 5; worksheets show who wrote what; self-assessment slips; exit ticket: talk parts.

SAFETY

- No practical hazards beyond normal classroom rules.
- Scissors for the panel cards: carry scissors with the points down.

WATCH OUT

- Own group's result is not 'the answer': class mean is better evidence; a word variable needs a bar chart.
- Time: have the results table ready; if short, do Panels 1, 4, 5 today and 6, 7 at the start of Week 10.

Language: Chinese talk welcome; panels in English or Chinese with English key words. Traps: pool 汇总, claim 主张, mean 平均值.

Next week: Keep posters flat, talk cards in folders and exit tickets; posters go up in talk order for the Week 10 Expo

PREP FOR ONE CLASS

before the lesson about 75 min (45 could be done by someone else); after about 51 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Book Week 9 printing with reprographics (part of the Week 7 booking) 5 min · teacher

THE WEEK BEFORE

- [ ] Print A3 posters, panel cards (100%), pooling sheets, slips, talk cards 10 min · reprographics
- [ ] Print worksheets (48 sheets) and kit card, from the language-pass designs 5 min · reprographics
- [ ] Cut self-assessment slips in half (24) and fold the 24 talk frame cards 10 min · technician
- [ ] Print and laminate sample data cards (3 per plan or only your plan's 3) 10 min · technician
- [ ] Write results table on A3 from Week 8 photo (Plan A: add melting rows) 15 min · teacher
- [ ] Choose a flat, safe place to store the A3 posters overnight 5 min · teacher

THE DAY BEFORE

- [ ] Put missing Week 8 results sheets in folders; pick groups for sample data 5 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Set out group kits, role and frame cards, sample cards, A3 table, slips 10 min · technician

AFTER THE LESSON

- [ ] Clean-up: posters flat in folders, slips collected, glue capped and counted 3 min · students
- [ ] Store posters flat; file worksheets and self-assessment slips 3 min · teacher
- [ ] Read every poster; pencil in draft levels for strands 1 to 5 in the tracker 30 min · teacher
- [ ] Compare slips with your judgement; note gaps of 2+ levels for a chat 10 min · teacher
- [ ] Read exit tickets: every group member has a talk part in poster order 5 min · teacher

PRINT

- Worksheet x24 (2 sheets each, double-sided)
- Pooling sheet x6 (1 per group, A4 landscape, double-sided)
- Pooling sheet side 1 enlarged to A3 x1 (board table)
- Poster template x7 (A3, single-sided; 6 groups plus 1 spare)
- Panel card set x7 (4 sheets each, single-sided, print at 100%)
- Self-assessment slip x24 (A5, 2 per sheet: 12 sheets, cut in half)
- Talk frame card x24 (1 sheet each, double-sided, folded)
- Sample data cards, laminated: Plan A x3 (double-sided), Plan B x3
- Teacher copies: kit card (A5, 2 sides), sample data answers (optional)

GET OR MAKE (NEW THIS LESSON)

- Laminating pouches A4 x6 (3 if only your plan's cards)
- Glue sticks x12 (2 per group)
- Thick black felt-tip pens x12 (2 per group)
- Thin black felt-tip pens x12 (2 per group)
- Class results table on the A3 board copy (make)

Have ready from earlier weeks: Group folders x6 (Week 7 and 8 sheets), Role card sets x6 (with language buddy), Sentence frame card sets x6, My Science Dictionary x24, Pencils x24, Scissors x6 (1 per group), 30 cm rulers x12 (2 per group), Logbooks x24 in class box (optional)